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ORIGINAL PAPER

Diffuse and specular component separation using multispectral and polarization images

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Abstract

Reflected light can be seen as a mixture of body reflection (diffuse component) and interface reflection (specular component). Applications using the diffuse reflection model assume Lambertian surfaces, and consider the specular reflection pixels as outliers. To recover or remove the highlights, a computational algorithm is used to separate the components in color space. In this paper, we generalize a polarization-based separation method to the case of multispectral images with K channels. A spectropolarimetric synthetic dataset of 10 scenes, including diffuse and specular reference images, is made available for both RGB and multispectral modalities. We found that our spectral adaptation qualitatively and quantitatively improves the separation results as compared to the RGB version. We also propose an algorithm optimization to reduce the computation time without sacrificing image quality.

Keywords Color and multi-spectral processing · Polarization imaging · Restoration · Reflection component separation

1 Introduction

Reflected light from an object can be seen as a combination of diffuse and specular reflection components, where the body reflection generates the diffuse component and the interface reflection generates the specular component [1]. Measurement of diffuse and specular reflection components in a scene is a challenging task that can help several imaging applications such as material inspection [2], highlight detection and removal [3, 4], illuminant estimation [5–7], and stereo reconstruction or rendering [8]. These applications need consistent surface information to perform well, because the presence of specularities can significantly affect the appearance of surfaces.

There are two main categories of algorithms able to separate specular and diffuse components: namely single-image and multiple-image methods. The single-image methods are mostly based on the analysis of the cluster formed by the data in the RGB space [9, 10], e.g. invoking the dichromatic [1] or the Cook and Torrance reflection models [11]. The second group of methods uses multiple images, either based on partial differential equations, an image sequence, or multiple-flash images. The reader may refer to two reviews from Artusi *et al.* [12] and Khan *et al.* [6] for a complete overview of these algorithms.

One type of method is to use the polarization information, where the specular component is assumed to be partially polarized [2, 13–16]. It uses a linear polarizer which is rotated in front of an RGB camera. For each pixel, rotating the polarizer produces a periodic intensity variation with two extrema [I_{min} and I_{max}] (see Figure 1). These maxima are used to separate the diffuse and specular components and find the illumination color in the RGB space. An example of separation result using an RGB and polarization method [13] is shown in Figure 2, where we reimplemented the method from the initial paper.

Polarization-based methods have been categorized as multiple image methods [12] for some time, because the original way of capturing the polarization information was to rotate a polarizer in front of a camera and to grab multiple images. But this categorization is no longer valid [17], since recent filter array sensors (e.g. Sony IMX250MYR or IMX253MYR) allow capturing linear polarization states in a snapshot way. In the same way, recent advances in Spectral Filter Arrays (SFA) enable the snapshot capture of multispectral images with $K > 3$ spectral channels [18, 19]. SFA sensors are commercially available from companies such as Silios Tech-

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nologies [20], Imec [21], or Spectral Devices [22]. The emergence of new sensors that simultaneously integrate multispectral and polarization information is leading to a new imaging paradigm, offering significant benefits for computer vision algorithms [23, 24].

To our knowledge, no method using multispectral and polarization information exists at the moment. When the color of the diffuse component is close to that of the specular one, the dichromatic reflection model leads to a bad separation result in the color space and therefore cannot be invoked with confidence [13]. We assume that doing the separation in a spectral space with $K > 3$ channels can help to: 1-estimate the spectral signature of illumination more precisely, 2-increase the number of pixels for which the assumption of the dichromatic model is valid and 3-stabilize the separation thanks to the increase in spectral resolution provided by spectral data.

This paper adapts and generalizes a color and polarization-based separation method to multispectral imaging. The mathematical formulation is redefined in K dimensions formed by the spectral sensor space, and the separation is solved using linear algebra. This has the advantage of being adaptable to any number of spectral channels and any wavelength range. We provide a synthetic dataset of spectropolarimetric images and evaluate on it. This could further help the design and evaluation of the separation algorithms by the community.

The paper is organized as follows. The reflection models relative to spectral and polarization imaging are defined in Section 2. Then, in Section 3, we provide a method to separate reflection components using multispectral and polarization information. An experimental protocol is given in Section 4. Results are presented and discussed in Section 5.

2 Background

2.1 Polarimetric imaging

At the macroscopic level, considering time-averaged values, the polarization state of light is commonly within the Stokes formalism [26]: we consider a wavelength-dependent four-component vector $\boldsymbol{S} = [S_0\ S_1\ S_2\ S_3]^T$ named Stokes vector. The total light intensity is S_0, whereas S_1 represents the intensity difference through 0° and 90° polarizers, and S_2 the intensity difference through 45° and 135° polarizers. S_3 refers to the circular component, i.e. the left or right-handed polarization component.

A linear Polarization State Analyzer (PSA), composed by a light sensing element coupled with a mere rotating linear polarizer, permits to estimate the first three Stokes components. The principle of a typical PSA is shown in Figure 1(a). The variation in intensity I relative to the polarizer angle θ follows a sinusoidal Malus law, as illustrated in Figure 1(b).

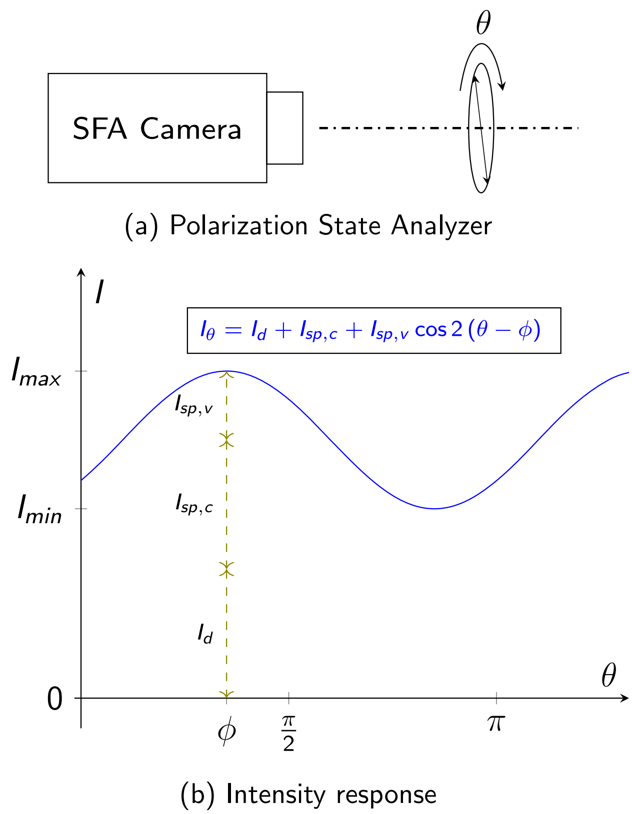


Fig. 1 **(a)** A spectropolarimeter composed by a Spectral Filter Array (SFA) camera and a linear polarizer. **(b)** After reflection, the light can become partially linearly polarized at a specific angle ϕ. Rotating a linear polarizer generates a phase-shifted sinusoid

It can be written as a linear relationship by:

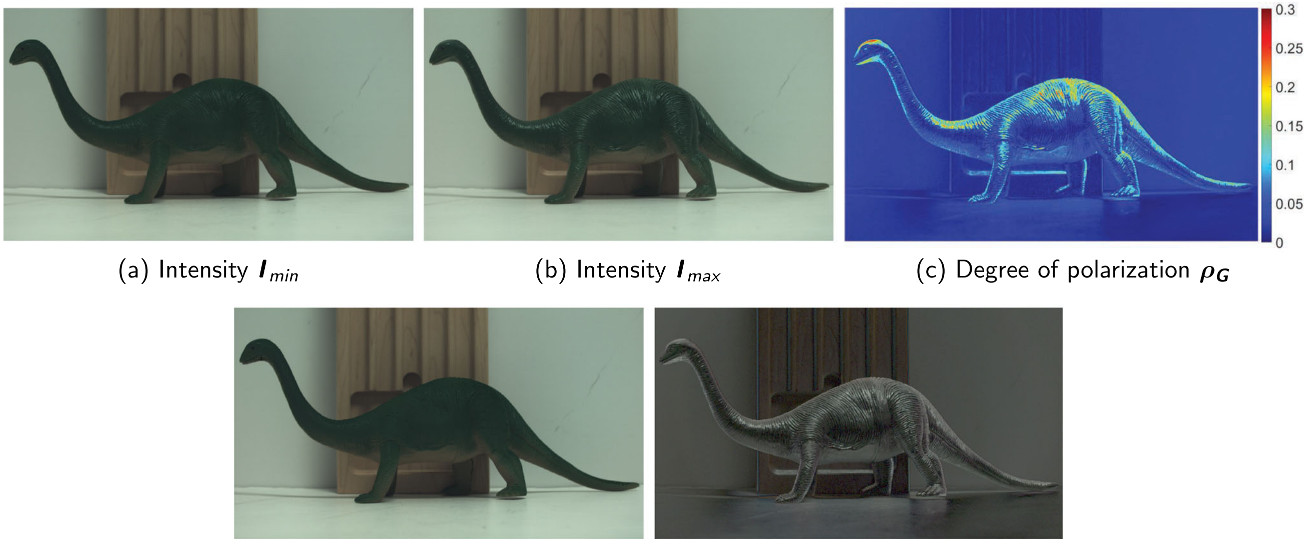
$$I_\theta = \boldsymbol{A}_\theta \boldsymbol{S}\ , \tag{1}$$

where I_θ is the pixel response at a particular angle θ, $\boldsymbol{S}$ is the polarization state of the input light, and $\boldsymbol{A} = \begin{bmatrix} a_0\ a_1\ a_2\ 0 \end{bmatrix}$ is the analyzer vector which embeds the polarization characteristics of the PSA: the transmission, the polarizing angle θ, and the extinction ratio coefficient.

In this work, we consider a linear PSA with $M = 4$ polarizer orientations at $\theta \in \{0°, 45°, 90°, 135°\}$. This is the most common configuration. Choosing $M = 3$ would have been sufficient, but $M = 4$ allows the user to lower noise [27]. The intensity responses of one pixel are:

$$\boldsymbol{I} = \begin{bmatrix} I_0 \\ I_{45} \\ I_{90} \\ I_{135} \end{bmatrix} = \mathbf{XS} = \frac{1}{2} \begin{bmatrix} 1 & 1 & 0 & 0 \\ 1 & 0 & 1 & 0 \\ 1 & -1 & 0 & 0 \\ 1 & 0 & -1 & 0 \end{bmatrix} \begin{bmatrix} S_0 \\ S_1 \\ S_2 \\ S_3 \end{bmatrix}\ , \tag{2}$$

where the polarimetric measurement matrix $\boldsymbol{X}$ is defined with the rows containing the four analyser vectors $\boldsymbol{A}_\theta$. In case of



(a) Intensity $\boldsymbol{I}_{min}$ (b) Intensity $\boldsymbol{I}_{max}$ (c) Degree of polarization $\boldsymbol{\rho}_G$

(d) Diffuse image $\boldsymbol{I}_d$ (e) Specular image $\boldsymbol{I}_{sp}$

Fig. 2 Reflection separation result using RGB and polarization. We reimplemented the method from Nayar *et al.* [13]. Scene from the dataset by Qiu *et al.* [25]. Only the green channel is shown for the degree of linear polarization image in (c). RGB specular image (e) is γ-corrected for visualization purpose ($\gamma = 2.2$)

non-idealities, the matrix $\boldsymbol{X}$ must be calibrated [28, 29]. In the following, the linear polarizer is considered as ideal.

Estimating the linear Stokes vector $\mathbf{S}$ is done at each pixel position of the sensor by $\mathbf{S} = \mathbf{X}^{+}\mathbf{I}$, where $\mathbf{X}^{+}$ is the pseudoinverse of $\mathbf{X}$. Our separation algorithm needs maximum and minimum intensity values I_{min} and I_{max}, which are evaluated over a π period of θ. These values have been estimated using a non-linear least-square fitting algorithm in the work that inspired this paper [13], but this method is very computationally intensive. Instead, we use the Stokes components to directly compute I_{min} and I_{max} [14]:

$$I_{min} = \frac{S_0 - \sqrt{S_1^2 + S_2^2}}{2} \qquad I_{max} = \frac{S_0 + \sqrt{S_1^2 + S_2^2}}{2} . \tag{3}$$

For a polarimeter with K spectral channels, I_{min} and I_{max} are estimated for each spectral channel $k \in \{1, ..., K\}$.

2.2 Reflection model

We explain here the principle involved in the separation method, that uses multispectral and polarization images.

2.2.1 Spectral considerations

The principle of the separation algorithm is based on the dichromatic reflection model [1, 10], which describes the reflectance function as the weighted sum of the specular reflectance (direct surface reflection) and the diffuse reflectance (sub-surface reflection). The intensity vector $\boldsymbol{I}$, formed by the K spectral channels can therefore be expressed as the sum of two intensity component vectors as follows:

$$\boldsymbol{I} = \boldsymbol{I}_d + \boldsymbol{I}_{sp} , \tag{4}$$

where $\boldsymbol{I}_d$ is the diffuse component vector and $\boldsymbol{I}_{sp}$ is the specular component vector.

The dichromatic reflection model also suggests that, for dielectrics, the spectral distribution of the diffuse component is linked to the material, while the spectral distribution of the specular component is that of the illumination. For a pixel to be separable, it is necessary that the directions of the vectors $\boldsymbol{I}_d$ and $\boldsymbol{I}_{sp}$ are significantly different.

2.2.2 Polarization considerations

The surface may be composed of several planar elements, or microfacets, where each facet has its own orientation which may differ from the overall larger-scale orientation. The result is a specular component that "spreads" around the main specular direction [30]. The light becomes partially polarized after reflection of unpolarized light, but it is assumed that the polarization of the diffuse component is negligible compared to the specular component. The rotation of a linear polarizer in front of a camera leads to an intensity measurement such that:

$$I_\theta = I_d + I_{sp,c} + I_{sp,v} \cos 2(\theta - \phi) , \tag{5}$$

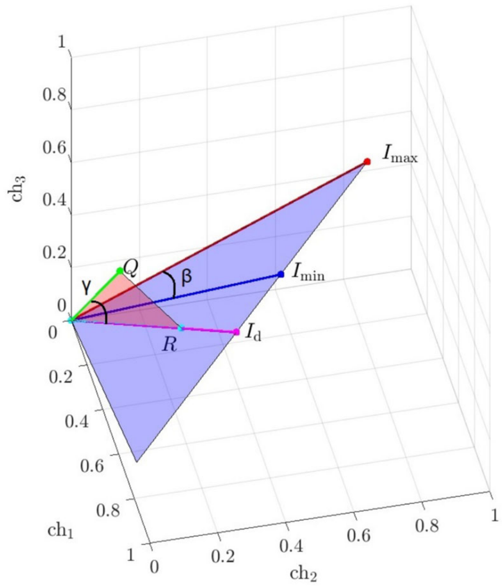


Fig. 3 A specular pixel varies in intensity along a line in the sensor space. The method takes a set of neighbour diffuse pixels $\boldsymbol{Q}$, verify that the spectra of $\boldsymbol{Q}$ are close to the plane ($\boldsymbol{O}$, $\boldsymbol{I}_{min}$, $\boldsymbol{I}_{max}$), and compute an average estimate of $\boldsymbol{I}_d$ from all projections of the neighbours. We generalize the principle in N-D space

where $I_{sp,c}$ is the constant specular component with respect to the polarization filter angle θ, $I_{sp,v}$ is the variable specular component, and ϕ is the phase of the sinusoid (i.e. the linear polarization angle of the light). A visualization of the intensity variation is shown in Figure 1(b). From Eq. 5, we see that separating the diffuse and specular components is an ill-posed problem, and is equivalent to estimating the constant specular component $I_{sp,c}$.

3 Method

Our method starts from a state-of-art method [13] that uses the color and polarization images captured from a scene to separate the diffuse and specular reflection components. Up to now, this method was limited to the RGB sensor space. We adapt the algorithm to the multispectral case, where more than three spectral bands are captured by a polarization and multispectral/hyperspectral acquisition setup. Our algorithm is not limited in the number of spectral channels, and can therefore be used to assess or optimize imaging systems to a specific separation application.

The principal assumptions of the algorithm are:

- Reflections follow a dichromatic model, where specular component is partially polarized and diffuse component is almost unpolarized. This means that the variation of the polarized intensity bounded by I_{min} and I_{max} is only due to the specular component.
- The spectrum of the light source is not that of the reflectance of the material, then the diffuse components and specular components do not share the same spectrum direction in the K-dimensional spectral channel space.
- We estimate the diffuse component using the neighbourhood of the pixel, using a set of available diffuse pixel values (or already been the subject of a previous separation).

The algorithm starts from only two spectral images with K spectral channels : $\boldsymbol{I}_{min}$ and $\boldsymbol{I}_{max}$, computed by Eq. 3.

3.1 Pixel labeling

The first step is to create a map where each pixel is labelled as specular or diffuse. The classification into specular or diffuse is done by using the method by Nayar *et al.* [13]. The Degree of Linear Polarization (DoLP) is computed by-pixel and by-channel using $\rho = \frac{I_{max}-I_{min}}{I_{max}+I_{min}}$. Then, a threshold is applied to check if the DoLP is sufficient to consider the pixel as specular :

$$\Omega_1 = 1 \text{ if } \max_{k\in\{1,\ldots,K\}} (\boldsymbol{\rho}_1, \boldsymbol{\rho}_2, \ldots, \boldsymbol{\rho}_K) > T_1 \text{ else } 0\ , \tag{6}$$

where Ω_1 is a binary map for specular pixels, and T_1 is a variable threshold defined as in [13]: T_1 ranges from 5% for points where I_{min} is greater than $\frac{200}{255}$, to about 10% when I_{min} is around $\frac{20}{255}$. This behavior considers the potential for noisy ρ values when I_{min} is low.

For pixels classified as specular, it is necessary to verify if the spectrum of the specular component is different from that of the diffuse one, i.e. if the Dichromatic reflection model holds. This is verified by computing the spectral angle (β in Figure 3) between $\boldsymbol{I}_{min}$ and $\boldsymbol{I}_{max}$ vectors from the origin, and verifying that the angle value is sufficient:

$$\Omega_2 = 1 \text{ if } \cos^{-1}\left(\frac{\boldsymbol{I}_{min}\boldsymbol{I}_{max}^T}{\|\boldsymbol{I}_{min}\|\|\boldsymbol{I}_{max}\|}\right) > T_2 \text{ else } 0\ , \tag{7}$$

where T_2 is a fixed threshold.

If the two previous tests Ω_1 and Ω_2 are true, we search for available diffuse pixels around the specular pixel of interest (next subsection).

3.2 Find diffuse neighbours

The second step is to consider the direct neighborhood of the pixel to be separated, and search for the neighbors for which the diffuse component is available (or previously calculated).

To this end, we define a local patch Ω, centered at the pixel of interest. We choose 15×15 pixels as an initial window size. We store the neighborhood values in $\boldsymbol{Q} = \boldsymbol{Q}_1, \boldsymbol{Q}_2, ..., \boldsymbol{Q}_J$, where J is a dynamic number of valid neighboors, initially set at $J = 15 \times 15$.

From $\boldsymbol{Q}$, we search for pixels that share a similar spectrum as the pixel of interest. In other words, we detect which points in $\boldsymbol{Q}$ are close to the plane $(\boldsymbol{O}, \boldsymbol{I}_{min}, \boldsymbol{I}_{max})$. This is done by computing the angle between $\boldsymbol{R}$ and $\boldsymbol{Q}$ (γ in Figure 3) as follows:

$$\gamma = \cos^{-1}\left(\frac{\boldsymbol{Q}\boldsymbol{R}^T}{\|\boldsymbol{Q}\|\|\boldsymbol{R}\|}\right), \tag{8}$$

where $\boldsymbol{R}$ is the projection of $\boldsymbol{Q}$ on plane $(\boldsymbol{O}, \boldsymbol{I}_{min}, \boldsymbol{I}_{max})$. The projection operation is defined later in Eq. 10. If the condition $\gamma < T_3$ is not verified (where T_3 is a choosen parameter), the neighbour value is omitted from $\boldsymbol{Q}$, decreasing the number J to $J-1$. The diffuse component computation (next section) is only done if the number of remaining neighbors is $J \geq T_4$.

3.3 Component Computation

From the selected neighbour pixels $\boldsymbol{Q}$ that satisfy the previous constraints, we compute all the estimations $\boldsymbol{P}$ of the diffuse component.

3.3.1 Mathematical background

We considering two full rank matrices $\boldsymbol{V}, \boldsymbol{W} \in \mathbb{R}^{n\times r}$ with $r < n$ such that $\det(\boldsymbol{W}^T\boldsymbol{V}) \neq 0$. The projection onto $\mathcal{V} = \mathrm{range}(\boldsymbol{V})$ orthogonally to $\mathcal{W} = \mathrm{range}(\boldsymbol{W})$ is given by the following matrix :

$$\Pi_{\mathcal{V},\mathcal{W}} = \boldsymbol{V}(\boldsymbol{W}^T\boldsymbol{V})^{-1}\boldsymbol{W}^T \;. \tag{9}$$

This matrix is invariant under any change of basis of $\mathcal{V}$ and $\mathcal{W}$. Indeed, considering $\boldsymbol{V} = \tilde{\boldsymbol{V}}\boldsymbol{P}$ and $\boldsymbol{W} = \tilde{\boldsymbol{W}}\boldsymbol{Q}$ with $\boldsymbol{P}, \boldsymbol{Q} \in \mathrm{GL}_r(\mathbb{R}) = \{M \in \mathbb{R}^{r\times r} \,|\, \det(M) \neq 0\}$ leads to:

$$\begin{aligned}
\Pi_{\mathcal{V},\mathcal{W}} &= \boldsymbol{V}(\boldsymbol{W}^T\boldsymbol{V})^{-1}\boldsymbol{W}^T \\
&= \tilde{\boldsymbol{V}}\boldsymbol{P}\left((\tilde{\boldsymbol{W}}\boldsymbol{Q})^T\tilde{\boldsymbol{V}}\boldsymbol{P}\right)^{-1}(\tilde{\boldsymbol{W}}\boldsymbol{Q})^T \\
&= \tilde{\boldsymbol{V}}\boldsymbol{P}\boldsymbol{P}^{-1}(\tilde{\boldsymbol{W}}^T\tilde{\boldsymbol{V}})^{-1}\boldsymbol{Q}^{-T}\boldsymbol{Q}^T\tilde{\boldsymbol{W}}^T \\
&= \tilde{\boldsymbol{V}}(\tilde{\boldsymbol{W}}^T\tilde{\boldsymbol{V}})^{-1}\tilde{\boldsymbol{W}}^T \;.
\end{aligned}$$

3.3.2 Diffuse Component Computation

For the present purpose, let $\mathcal{V} = \mathrm{range}(\boldsymbol{V})$, with $\boldsymbol{V} = [\boldsymbol{I}_{\min}\ \boldsymbol{I}_{\max}]$, be the linear span of the set $\{\boldsymbol{I}_{\min}, \boldsymbol{I}_{\max}\}$. The orthogonal projection of one specific neighbour $\boldsymbol{Q}$ on $\mathcal{V}$ denoted $\boldsymbol{R}$ is given by

$$\boldsymbol{R} = \Pi_{\mathcal{V},\mathcal{V}}(\boldsymbol{Q}) = \boldsymbol{V}(\boldsymbol{V}^T\boldsymbol{V})^{-1}\boldsymbol{V}^T\boldsymbol{Q} \;. \tag{10}$$

A diffuse candidate $\boldsymbol{P}$ is computed through the oblique projection of $\boldsymbol{I}_{\min}$ (or equivalently $\boldsymbol{I}_{\max}$) on $\mathcal{R} = \mathrm{range}(\boldsymbol{R})$ along $\boldsymbol{U} = \boldsymbol{I}_{\max} - \boldsymbol{I}_{\min}$. To do that, let us consider the vector $\boldsymbol{W} = \Pi_{\mathcal{V},\mathcal{V}}(\boldsymbol{X})$ where $\boldsymbol{X}$ is any non-zero vector orthogonal to $\boldsymbol{U}$. Thus, the diffuse component $\boldsymbol{P}$ is obtained through the projection of $\boldsymbol{I}_{\min}$ on $\mathcal{R}$ orthogonally to $\mathcal{W} = \mathrm{range}(\boldsymbol{W})$.

$$\boldsymbol{P} = \Pi_{\mathcal{R},\mathcal{W}}(\boldsymbol{I}_{\min}) = \boldsymbol{R}(\boldsymbol{W}^T\boldsymbol{R})^{-1}\boldsymbol{W}^T\boldsymbol{I}_{\min} \;. \tag{11}$$

It should be noted that, since $\boldsymbol{U} \in \ker(\Pi_{\mathcal{R},\mathcal{W}})$, $\boldsymbol{P} = \Pi_{\mathcal{R},\mathcal{W}}(\boldsymbol{I}_{\max})$.

For each candidate $\boldsymbol{P}$, we compute its distance to $\boldsymbol{I}_{\min}$ by $d = \|\boldsymbol{I}_{\min} - \boldsymbol{P}\|$. We verify that this distance is positive, i.e. each estimation $\boldsymbol{P}$ is below $\boldsymbol{I}_{\min}$. Then, we compute the mean distance $\overline{d}$ over all the selected neighbors, along with the standard deviation $std(\boldsymbol{d})$. If $std(\boldsymbol{d}) < T_5$, the estimation is accepted and we finally compute $\boldsymbol{I}_d$ by [13]:

$$\boldsymbol{I}_{\mathrm{d}} = \boldsymbol{I}_{\min} - \overline{d}\frac{\boldsymbol{I}_{sv}}{\|\boldsymbol{I}_{sv}\|}. \tag{12}$$

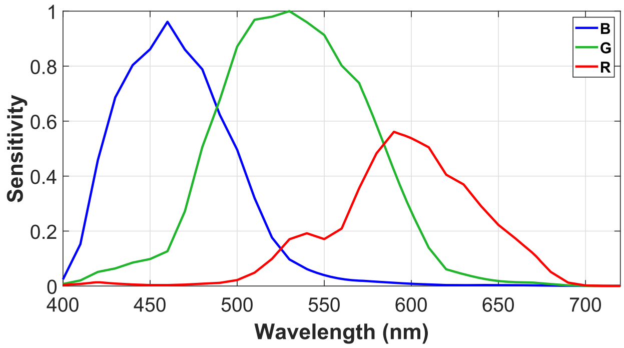
It should be noted, that for obtaining $\boldsymbol{I}_{sp}$, we can simply substract $\boldsymbol{I}_d$ from $\boldsymbol{I}$ (see Eq. 4).

4 Experiment

4.1 Synthetic dataset

To evaluate quantitatively and qualitatively the separation results, we generate a spectropolarimetric synthetic dataset of 10 scenes using the rendering system Mitsuba 3 [31], which use was already reported for polarization rendering [32]. We generated synthetic RGB ($K = 3$ spectral channels) and multispectral ($K = 9$) scenes, using measured camera sensitivities as input for the simulation. We use synthetic objects from [33], and adapt their scene generation methodology through the following steps.:

- The spectral sensitivities are set to those of the Canon EOS-1D Mark III [34] for RGB, and those of the 4.2MP Silios CMS4-V spectral filter array camera for multispectral (see Figure 4). The multispectral camera captures $K = 9$ spectral channels referred to as ch_{1-9}: 8 bands centered on 430, 460, 500, 530, 570, 610, 650, and 690 nm, and one panchromatic band in the visible spectrum.



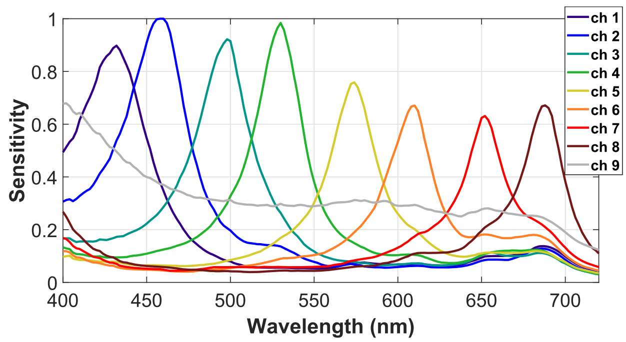


Fig. 4 **(a)** Spectral sensitivities of the Canon EOS-1D Mark III and the **(b)** 4.2MP Silios CMS4-V multispectral camera used in generation of the synthetic dataset

- A linear synthetic polarizer is added in front of each camera system, to select the polarization at $\theta = \{0°, 45°, 90°, 135°\}$, as in Eq. 1.
- The BSDFs (Bidirectional Scattering Distribution Function) of the objects have been changed to *pplastic*, a polarization-compatible model in Mitsuba 3 that accounts for polarization reflection effects. We also modulate the specular reflection component (i.e. by setting the *specular_reflectance* parameter to zero) to generate purely diffuse reference images ($\boldsymbol{I}_{d,ref}$), used as ground-truths in our evaluation.

The data are linear high-dynamic range images (*.exr* file format), and generated using the variant named *cuda_spectral_polarized_double* in Mitsuba 3. The sampler called *independent* is used to ensure equal noise distribution among all generated images.

A color version of the dataset, along with several examples of synthesized images, is provided in the supplementary information attached to this paper

The dataset will be made available once accepted.

4.2 Evaluation

Here, we qualitatively and quantitatively evaluate the robustness of the separation in the spectral space as compared to the RGB space. For both cases, we select the same thresholds as follows: T_1 as defined in subsection 3.1, $T_2 = 0.08$, $T_3 = 0.2$ (doubling every 10 iterations), $T_4 = 3$, and $T_5 = 0.14$. The neighborhood size is initially set at 15×15 pixels (in Section 3.2). We stop the algorithm after 30 iterations, assuming that this is enough for the algorithm to converge for both spaces.

To assess the performance, we apply two full-reference metrics between the estimated diffuse image $\boldsymbol{I}_d$ and the diffuse reference image $\boldsymbol{I}_{d,ref}$. We use PSNR (Peak Signal to Noise Ratio) as an objective quality metric, and SSIM (Structural Similarity Index Measure) [35] to be more sensitive to local structures and contrast changes. The computation of all PSNR/SSIM values is done for each spectral channel.

Using the initial parameters, the computation time is relatively important for this kind of algorithm, especially on non-parralelizable architectures. For optimization purpose, we evaluate the performance of the algorithm by varying the neighborhood size that is used for searching neighbour diffuse pixels. This is the parameter with the greatest impact on the number of operations in the algorithm.

5 Results and analysis

Figure 5 reports the visual results for two scenes and for both scenarios (RGB or multispectral). In subfigures 5l, 5n, 5p and 5r, we focus on two zones for which the estimation of the specular component was not entirely successful, especially in RGB. For both scenarios, a small part of specular pixels remain after separation. This can be explained by several reasons. First, the algorithm does not converge because there are not enough available diffuse neighbors, which does not respect the condition on the T_4 threshold, leaving the pixel at its initial value (replaced by $\boldsymbol{I}_{min}$ in our case). Secondly, a specular pixel may not be identified as such due to its low degree of polarization. This is a physical property coming from Fresnel coefficients [36], where the degree of polarization of the specular reflection component depends on the viewing angle and the refraction index. Under some conditions, a specular reflection may not be sufficiently polarized for the algorithm to work. It is to note that the residual specular intensity is much less pronounced in the multispectral image than in the RGB image (especially in the upper left corner of the ball in Figure 5d), which proves a better reconstruction.

Furthermore, subFigures 5h and 5j show that the multispectral case is less prone to artifacts and reconstruction

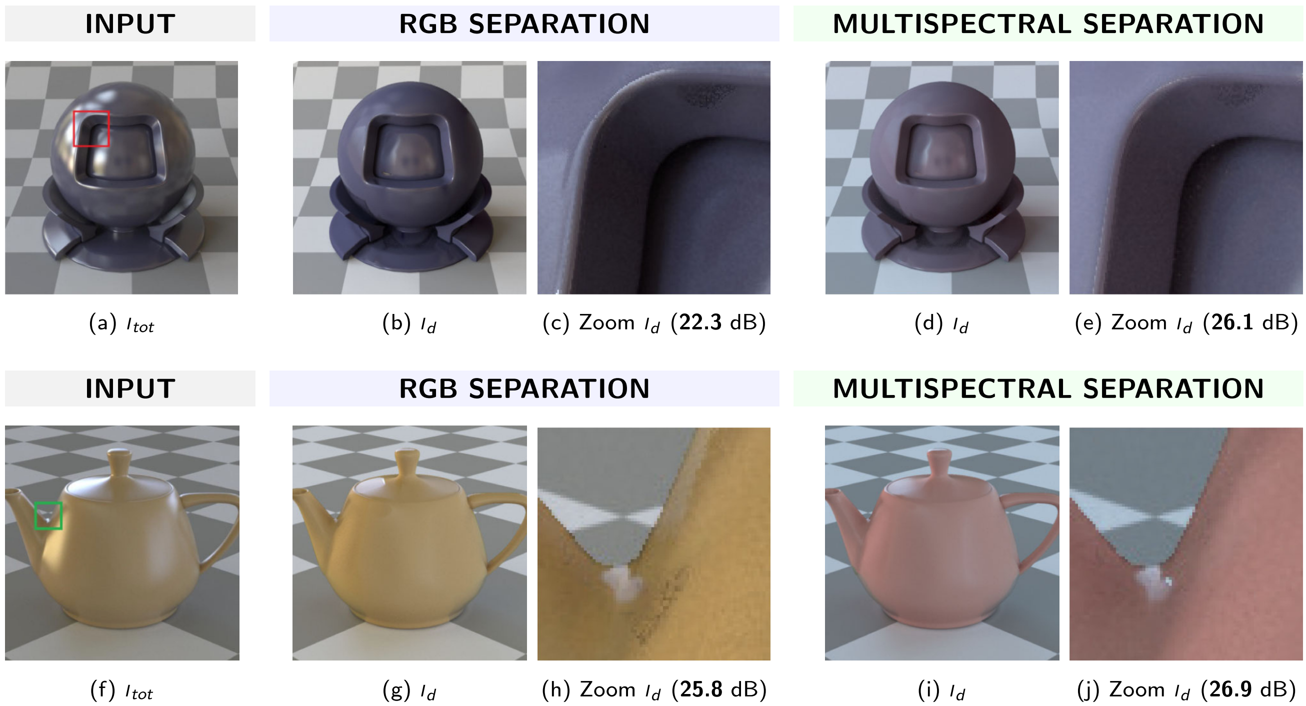


Fig. 5 Comparison of separation results for the *matpreview* and *teapot* scenes. The results show that the multispectral (MS) approach better recovers the diffuse component compared to the RGB version

noise: several dark spots appear below the remaining specularity area in 5h, but they are cancelled in 5j.

Figure 6 reports quantitative results for the two metrics used, PSNR and SSIM. A higher score is better for PSNR and SSIM. All color channels were considered ($K = 3$ for RGB, $K = 9$ for multispectral). The figure consists of boxplots describing the metrics distribution. The algorithm is run choosing a 15×15 neighborhood around each pixel. We can see that the scenario using multispectral data is giving better results by a small amount, along with reduced standard deviations. As compared to the RGB version, the average PSNR for multispectral images is increased by $0.54dB$, whereas average PSNR increases by 0.005. The standard deviations are roughly the same for PSNR, but decreased by 0.008 for SSIM results. This is a relatively small improvement, but this must be put into perspective with the fact that the total number of specular pixels to be reconstructed in the dataset is low compared to the pixels already considered as diffuse. This may explain this small increase in quality. When we consider subimages with a significant proportion of specular pixels, the improvement is very noticeable, as reported in subFigures 5c, 5e, 5h and 5j, with an improvement in PSNR up to $3.8dB$.

Figure 7 illustrates the results of the optimization procedure. We vary the size of the neighborhood considered when searching for diffuse pixels around the pixel of interest. Each point in the graphs corresponds to a complete evaluation of the algorithm on the entire database. Only the neighborhood size parameter is changed. The computing architecture is an Intel Xeon CPU E5-2690.

As a general comment, the computation time is relatively high for this processor-based computing architecture, i.e. around $500s$ for a 6×6 neighborhood. We can see that, for PSNR and SSIM metrics, there is a rapid convergence, and little enhancement is provided by using a window size larger than 8. It can also be noted that the difference in computation times between the RGB case and the spectral case is not proportional to the number of channels. For the same neighborhood, there is a relatively small gap between the computation time for RGB and Multispectral, approximatively a factor of 1.2.

It is important to note that the processor's sequential nature can significantly impact computation time. However, the time increase observed with larger windows would likely be less pronounced on a parallel processing system. This needs to be considered for future optimizations of the algorithm.

6 Conclusion

We present an extended color and polarization separation technique for multispectral imaging. By dimensioning the problem into an K-dimensional spectral sensor space, we develop a linear algebra-based solution that is adaptable to any number of spectral channels. We use and provide a synthetic spectropolarimetric dataset for benchmark and make

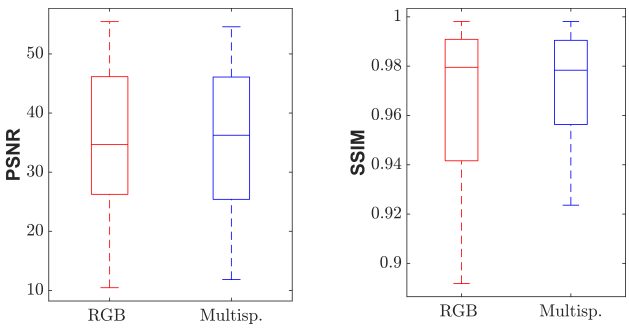


Fig. 6 Box plot of separation results computed on average data over all channels of all images in the database, and for each method. The central mark indicates the median. The boxes extend from the 1st quartile to the 3rd quartile, and the whiskers indicate the full ranges of the data

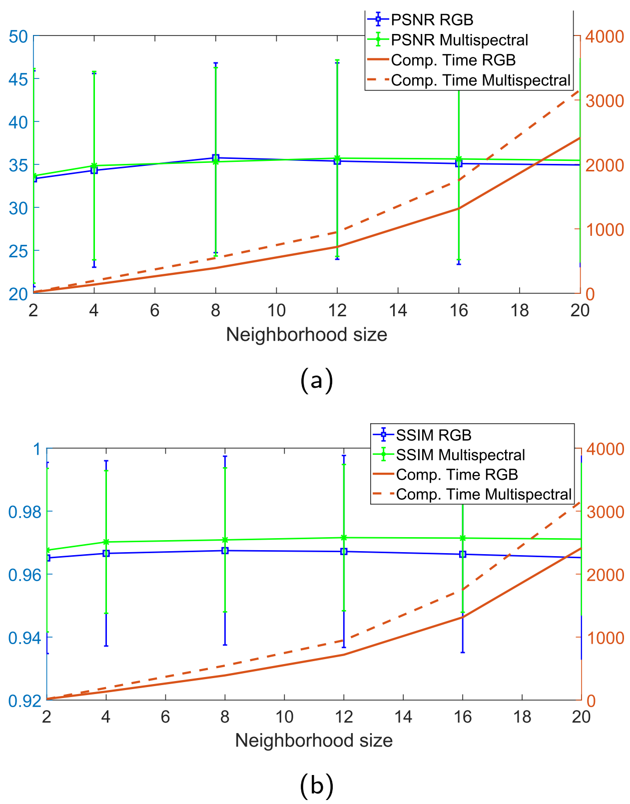


Fig. 7 Optimization results relatively to the neighborhood size that is considered when searching diffuse pixels around the pixel of interest

our separation algorithm's code publicly available. We found that our algorithm using 9 channels instead of 3 gives a rather small improvement on our database compared to the RGB case when entire images are considered. But when zooming in, we can see that specular pixels are clearly better separated from diffuse pixels, which is also sensible in metrics (up to 3.8 dB raise on PSNR).

For optimal computational efficiency, our findings indicate that a neighborhood size of 8 × 8 pixels is sufficient, as larger sizes do not provide significantly better PSNR and SSIM results in our evaluation framework. This parameter setting reduces the computation times by 5 for our target architecture compared to the initial implementation.

As a future work, we plan to exploit other wavelength ranges, from visible to SWIR, and evaluate the effect of spatial and spectral resolutions of images over the results, e.g. using mosaiced images from an hyperspectral sensor as input. We also plan to adapt other alternative separation algorithms specifically for multispectral data for benchmarking.

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Author contributions P.L. conceptualized the study and wrote the main manuscript text. P.L and B.M. designed the experiments and contributed to the development of the framework. P.L., B.M., and L.B performed the analysis and contributed to the result interpretation. All authors participated to prepare the figures. All authors reviewed and approved the manuscript.

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Data availability The dataset and code supporting the results of this study is available at https://doi.org/10.6084/m9.figshare.28151753

Declarations

Conflict of interests The authors declare that they have no conflict of interest.

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